The predual of bounded Schatten Schur multipliers is a Banach algebra under matrix multiplication

Autonomous mathematical discovery run `fa_banach_001`

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Status. Candidate full affirmative answer, likely valid pending expert review.

1 Source question

Let I be a set and $1 \leq p < \infty$. Arhancet defines $\mathfrak{R}_p^I$ as the natural predual of the bounded Schur multipliers on $\mathcal{S}_p^I$. Concretely, $\mathfrak{R}_p^I$ is the image of

$$\mathcal{S}_p^I \widehat{\otimes}_\pi \mathcal{S}_{p'}^I \longrightarrow \mathcal{S}_1^I, \qquad A \otimes B \longmapsto A * B,$$

with its quotient norm. Remark 3.8 asks whether $\mathfrak{R}_p^I$, equipped with ordinary matrix multiplication, is a Banach algebra [1, Remark 3.8, PDF p. 18].

> 18 Cédric Arhancet
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> *Remark* 3.8. We do not know if the space $\mathfrak{R}_p^I$ equipped with the usual matricial product is a Banach algebra. The Banach space analogue of Proposition 3.5 is false. It is the reason which explains that the method does not work for $\mathfrak{R}_p^I$. However, note that if $\mathfrak{M}_p^I = \mathfrak{M}_{p,cb}^I$ isometrically we have $\mathfrak{R}_p^I = \mathfrak{R}_{p,cb}^I$ isometrically. For $1 < p < \infty$, $p \neq 2$ the equality $\mathfrak{M}_p^I = \mathfrak{M}_{p,cb}^I$ is a classical open question.

We answer this affirmatively, with multiplication constant one. The point is that one only needs to amplify the very special tensors arising from two Hadamard factorizations. A rank-one decomposition controls those tensors directly, even though the general Banach-space amplification used in the source is false.

2 Preliminaries from the source

Write $A * B = (a_{ij} b_{ij})$. The quotient description gives

$$\|C\|_{\mathfrak{R}_p^I} = \inf \left\{ \sum_n \|A_n\|_p \|B_n\|_{p'} : C = \sum_n A_n * B_n \right\}. \tag{1}$$

The series in this formula is absolutely convergent in the projective tensor-product sense. The canonical inclusion $\mathfrak{R}_p^I \hookrightarrow \mathcal{S}_1^I$ is contractive. For $1 < p < \infty$, the identity on matrices is an isometric identification

$$\mathfrak{R}_p^I = \mathfrak{R}_{p'}^I; \tag{2}$$

these facts are recorded in Section 2 of [1].

3 Idea of the proof

Start with two elementary quotient representatives $A*B$ and $X*Y$. Their ordinary matrix product is a sum over the middle index k. For each k, put the corresponding column–row outer products into rank-one matrices P_k and Q_k; the kth summand is then exactly $P_k * Q_k$. When $p \geq 2$, the inclusion $\mathcal{S}^I_{p'} \subseteq \mathcal{S}^I_2$ turns the total projective cost of these rank-one factorizations into a Cauchy–Schwarz estimate with constant one. Summing this elementary estimate over arbitrary quotient representations proves submultiplicativity. The source isometry $\mathfrak{R}^I_p = \mathfrak{R}^I_{p'}$ transfers the result to $1 < p < 2$, while the source's completely bounded endpoint theorem handles $p = 1$.

4 The elementary product estimate

Lemma 1. *Let $2 \leq p < \infty$. For $A, X \in \mathcal{S}^I_p$ and $B, Y \in \mathcal{S}^I_{p'}$,*

$$(A * B)(X * Y) \in \mathfrak{R}^I_p$$

and

$$\|(A * B)(X * Y)\|_{\mathfrak{R}^I_p} \leq \|A\|_p \|B\|_{p'} \|X\|_p \|Y\|_{p'}. \tag{3}$$

Proof. For $k \in I$, define the rank-one matrices

$$P_k = (a_{ik} x_{kj})_{i,j \in I}, \qquad Q_k = (b_{ik} y_{kj})_{i,j \in I}.$$

Thus P_k is the outer product of column k of A and row k of X, and similarly for Q_k. Every Schatten norm of a rank-one matrix is the product of the two Euclidean norms, so, with

$$\alpha_k = \|A e_k\|_2, \quad \gamma_k = \|X^* e_k\|_2, \quad \beta_k = \|B e_k\|_2, \quad \delta_k = \|Y^* e_k\|_2,$$

we have

$$\|P_k\|_p = \alpha_k \gamma_k, \qquad \|Q_k\|_{p'} = \beta_k \delta_k.$$

Because $p' \leq 2$, the inclusion $\mathcal{S}^I_{p'} \subseteq \mathcal{S}^I_2$ is contractive. Hence Cauchy–Schwarz gives

$$\begin{aligned}
\sum_k \|P_k\|_p \|Q_k\|_{p'} &= \sum_k \alpha_k \gamma_k \beta_k \delta_k \\
&\leq \big(\sup_k \alpha_k \gamma_k\big) \sum_k \beta_k \delta_k \\
&\leq \|A\|_\infty \|X\|_\infty \|B\|_2 \|Y\|_2 \\
&\leq \|A\|_p \|X\|_p \|B\|_{p'} \|Y\|_{p'}.
\end{aligned}$$

Therefore $\sum_k P_k \otimes Q_k$ is absolutely convergent in $\mathcal{S}^I_p \widehat{\otimes}_\pi \mathcal{S}^I_{p'}$. Applying the Hadamard-product quotient map and comparing entries yields

$$\sum_k P_k * Q_k = \left(\sum_k a_{ik} b_{ik} x_{kj} y_{kj} \right)_{i,j} = (A * B)(X * Y).$$

The norm formula (1) now gives (3). $\square$

5 Full answer

Theorem 2. *For every set I and every $1 \le p < \infty$, $\mathfrak{R}_p^I$ is a contractive Banach algebra under ordinary matrix multiplication. Explicitly, for all $C, D \in \mathfrak{R}_p^I$,*

$$CD \in \mathfrak{R}_p^I, \qquad \|CD\|_{\mathfrak{R}_p^I} \le \|C\|_{\mathfrak{R}_p^I} \|D\|_{\mathfrak{R}_p^I}. \tag{4}$$

Moreover, for every $i, j \in I$, the net $\sum_{k \in J} c_{ik} d_{kj}$ over finite subsets $J \subset I$ converges absolutely to $(CD)_{ij}$.

Proof. First suppose $2 \le p < \infty$. Choose representations

$$C = \sum_n A_n * B_n, \qquad D = \sum_m X_m * Y_m$$

as in (1). Lemma 1 shows that the double series

$$\sum_{n,m} (A_n * B_n)(X_m * Y_m)$$

converges absolutely in $\mathfrak{R}_p^I$, with norm at most

$$\left(\sum_n \|A_n\|_p \|B_n\|_{p'} \right) \left(\sum_m \|X_m\|_p \|Y_m\|_{p'} \right).$$

The inclusion $\mathfrak{R}_p^I \hookrightarrow \mathcal{S}_1^I$ is contractive, and $\mathcal{S}_1^I$ is a Banach algebra under operator composition. Consequently the sum is exactly CD. Taking infima over the two representations proves (4).

If $1 < p < 2$, then $p' > 2$ and the isometry (2) transfers the already proved estimate from $\mathfrak{R}_{p'}^I$ to $\mathfrak{R}_p^I$. For $p = 1$, the source records $\mathfrak{R}_1^I = \mathfrak{R}_{1,cb}^I$ isometrically, and its Theorem 3.7 already proves the contractive multiplication estimate. This completes all values of p.

Finally, since $C, D \in \mathcal{S}_1^I \subset \mathcal{S}_2^I$, every row of C and every column of D lies in $\ell_2(I)$. Thus

$$\sum_k |c_{ik} d_{kj}| \le \left(\sum_k |c_{ik}|^2 \right)^{1/2} \left(\sum_k |d_{kj}|^2 \right)^{1/2} < \infty,$$

which proves the stated coordinate convergence. □

6 Scope, verification, and novelty

Theorem 2 fully answers the first question in Remark 3.8 for all exponents and all index sets. It does not settle the companion question whether every bounded Schur multiplier on $\mathcal{S}_p$ is completely bounded. The proof deliberately avoids that stronger conjecture.

The accompanying script in `code/verify_factorization.py` verifies, on random finite matrices and several exponents, the rank-one identity and every numerical inequality in Lemma 1. These computations are regression checks; the proof itself is exact and dimension-free.

A bounded novelty search, refreshed on 21 August 2026, covered the four run indexes, exact title and arXiv-id searches, exact phrases from Remark 3.8, and later work on the complete bounds of Schatten Schur multipliers. No later answer to the $\mathfrak{R}_p$ matrix-product question was found. The later literature found continues to study the distinct companion equality $\mathfrak{M}_p = \mathfrak{M}_{p,cb}$.

Human review recommendation. Check the four-factor rank-one decomposition in Lemma 1, especially the use of $\mathcal{S}_{p'} \subset \mathcal{S}_2$ for $p \geq 2$, and then check that the projective representations pass to the operator product through the contractive inclusion $\mathfrak{R}_p^I \hookrightarrow \mathcal{S}_1^I$.

References

[1] C. Arhancet, *Noncommutative Figà-Talamanca–Herz algebras for Schur multipliers*, Integral Equations Operator Theory **70** (2011), 485–510; arXiv:0911.4304v2.